

International Islamic University Chittagong
Department of Electrical and Electronic Engineering
Complex Engineering Problem

Course Title: Power System Analysis Sessional
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Project Title: Load flow analysis using any suitable load flow method

Introduction: Load flow analysis is an essential study in power system engineering that evaluates the steady-state behavior of an electrical network under given load conditions. It determines bus voltages, power flows, and system losses, which are vital for ensuring reliable and efficient operation. A multi-bus power system consists of interconnected buses classified as PQ (load), PV (generator), and slack buses, each with defined operating characteristics. This project applies load flow analysis to a multi-bus network to assess voltage performance, power distribution, and overall system stability.

Methodology: The load flow analysis starts with modeling the multi-bus power system using a single-line diagram that includes buses, generators, loads, and transmission lines. Each bus is classified as a slack, PV, or PQ bus based on known system variables. Required data such as line impedances, load demands, and generation limits are then collected to form the bus admittance matrix (Y-bus). A suitable numerical technique—such as Gauss-Seidel, Newton-Raphson, or Fast Decoupled—is applied to solve the nonlinear load flow equations. The solution is obtained iteratively until convergence is achieved, after which voltage profiles, power flows, and system losses are evaluated. Software tools like MATLAB or PowerWorld are used to improve accuracy and computational efficiency.

Assumptions for Load Flow Analysis: The network is an IEEE 5-bus system with- 2 Generator buses (Slack and PV types), 3 Load buses (PQ type), 7 Transmission lines connecting the buses. Specific values for bus and line data are used based on the Matric ID of one of our team members (ET233042) as follows

Bus Data: According to project requirement Total power demand in buses needs to be satisfied as $P_L = [PL1 \ PL2 \ PL3]$ MW and $Q_L = [QL1 \ QL2 \ QL3]$ MVAR where,

$PL1 = \text{Summation of last 3 digits of your roll number (ID) MW} = 6$

$PL2 = \text{Summation of last 3 digits of your roll number (ID) MW} + 50\text{MW} = 56$

$PL3 = \text{Summation of last 3 digits of your roll number (ID) MW} + 50\text{MW} = 56$

$QL1 = \text{Summation of last 2 digits of your roll number (ID)} \times 5 \text{ MVAR} + 20\text{MVAR} = 50$

$QL2 = \text{Summation of last 2 digits of your roll number (ID)} \times 5 \text{ MVAR} + 30\text{MVAR} = 60$

$QL3 = \text{Summation of last 2 digits of your roll number (ID)} \times 5 \text{ MVAR} + 40 \text{ MVAR} = 70$

Assuming other values the bus data are given in the table below

Table 1 Bus Data

Bus No:	Type	Voltage (PU)	Generation		Load	
			MW	MVAR	MW	MVAR
1	Slack	1.06	0	0	0	0
2	PV	1	80	110	12	10
3	PQ	1	0	0	6	50
4	PQ	1	0	0	56	60
5	PQ	1	0	0	56	70

Line Data: Seven transmission line are present in this system. Since no data are given for line resistance and reactance let assume the data using the **last two digit of our student's ID's (42, 53, 67, 72)** for line resistance and double it for line reactance. Assuming other values the line data for IEEE five bus system are given in the table below.

Table 2 Line Data for IEEE 5 Bus System

Line		Line Resistance	Line reactance	Line Charging Susceptance
From	To	R(PU)	X(PU)	B
1	2	0.04	0.08	j0.06
1	3	0.02	0.04	j0.05
2	3	0.05	0.1	j0.04
2	4	0.03	0.18	j0.04
2	5	0.06	0.12	j0.03
3	4	0.07	0.14	j0.02
4	5	0.02	0.04	j0.05

Steps: The simulation has been performed using Power World Simulator maintaining the following steps-

- 1) System components (Bus, Load, Generator, Lines) were placed using the graphical interface.
- 2) All the bus data, load, generation limits, and line impedances were configured.
- 3) Load Flow Analysis was performed through Run Mode to obtain system voltages, power flows and losses.

Model of the network:

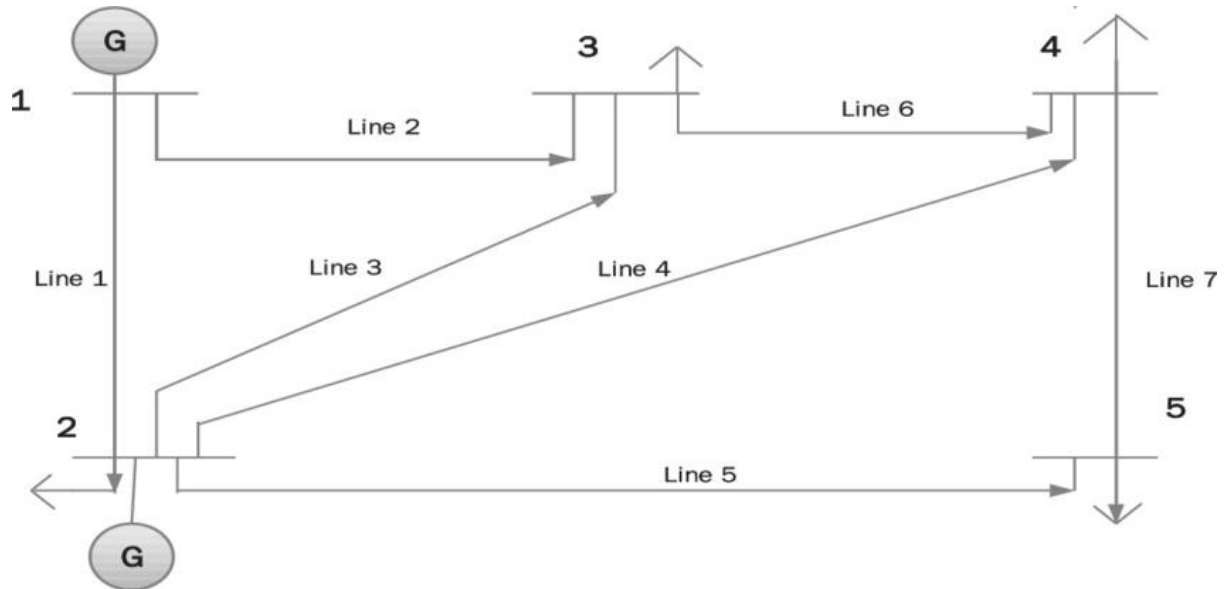


Figure 1 Schematic of IEEE 5-bus system

System Network Simulation:

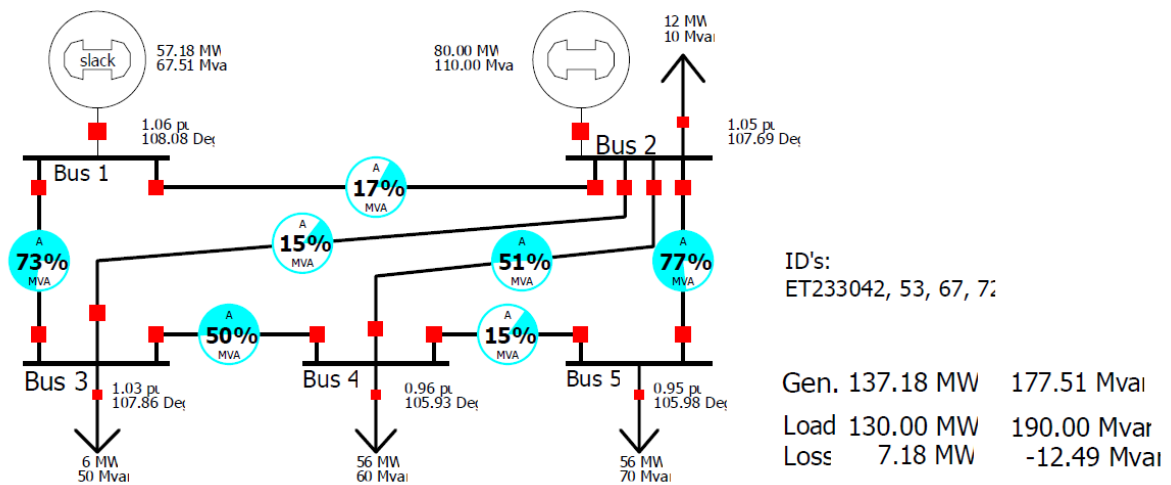
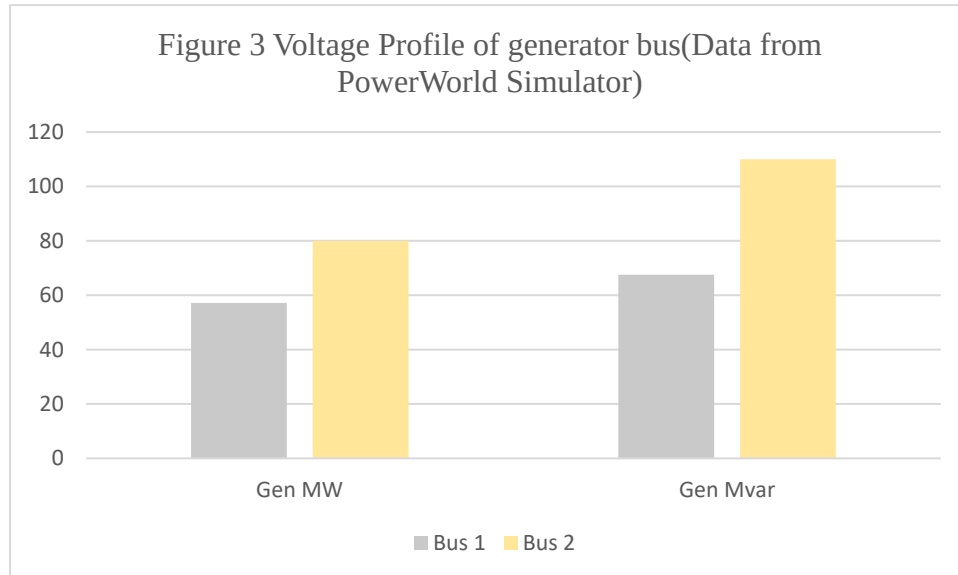


Figure 2 IEEE 5-Bus System Network Simulation Result.

Data Table-01: Bus Voltage Profile of IEEE 5-Bus System

Name	Nominal kV	PU Volt	Volt (kV)	Angle (Deg)	Gen MW	Gen Mvar	Load MW	Load Mvar
Bus 1	138	1.06	146.28	108.08	57.18	67.51		
Bus 2	138	1.04596	144.343	107.69	80	110	12	10
Bus 3	138	1.02844	141.925	107.86			6	50
Bus 4	138	0.95828	132.243	105.93			56	60
Bus 5	138	0.95029	131.14	105.98			56	70

Graphical Results:



Results: Total Generation $\approx$ 138 MW and 178 MVar, where the slack bus supplies the deficit power to maintain system balance.

Total Load = $(12 + 6 + 56 + 56) = 130$ MW and $(10 + 50 + 60 + 70) = 190$ MVar.

Total System Losses $\approx$ 8 MW and a significant amount of reactive power losses due to high inductive loading and line reactance.

Due to the very high reactive power demand specified by the system data, considerable voltage drops are observed at the PQ buses. Although the load flow solution converges successfully, such operating conditions would lead to voltage instability if applied in a real-world power system without adequate reactive power compensation.

Importance of Stability for Societal Energy Demands: Power system stability is vital for delivering a continuous and reliable electricity supply in response to increasing societal energy demands. A stable power system minimizes the risk of blackouts, safeguards critical infrastructure, and supports economic activities. Since modern society relies heavily on uninterrupted electricity for healthcare, communication, transportation, and industry, maintaining system stability is essential for public safety, economic development, and overall quality of life.

Recommendations:

- 1) **Reactive Power Support:** Install capacitor banks at critical buses to enhance local voltage profiles and maintain voltage stability.
- 2) **Generator Dispatch Optimization:** Optimize generator output by increasing generation at lightly loaded buses to reduce stress on heavily loaded buses and improve overall system balance.

- 3) Network Load Balancing: Reassess and balance power flow across parallel transmission lines to minimize losses and prevent line overloading.

Justification for CEP Based on Washington Accord Standards: Based on the simulation results and tabulated data, this assignment clearly qualifies as a Complex Engineering Problem (CEP) and involves Complex Engineering Activities (CEA) in accordance with the Washington Accord standards. The project demonstrates compliance with these standards through the application of advanced power system analysis techniques, effective use of modern simulation tools, and systematic evaluation of technical challenges and their real-world impacts.

Complex Engineering Problem (CEP) Standards:

P1: Depth of Knowledge - Applied advanced load flow analysis using the Newton–Raphson method in PowerWorld.

P2: Technical Challenges - Managed voltage stability, power flow, and loss minimization while maintaining voltage limits.

P5: Societal & Environmental Impact - Analyzed system losses to demonstrate efficient power delivery, reliability, and reduced energy wastage.

CEA – Complex Engineering Activities

- A1: Modeled IEEE 5-bus system with slack, PV, PQ buses, generation, loads, and transmission.
- A3: Diagnosed and fixed power transfer issues; fine-tuned parameters and validated results.
- A5: Used professional simulation software and presented results with standardized diagrams and structured data, following industry documentation practices.

Discussion: The load flow study showed that several bus voltages were outside the acceptable range, signaling potential system instability and the need for corrective measures. Notable active and reactive power losses were detected, which can reduce efficiency and raise operating costs. To address these issues, strategies like reactive power support, optimized generator dispatch, and enhanced network design—including equipment upgrades and reconfiguration are recommended to improve overall performance and reliability.

Conclusion: Load flow analysis is essential for assessing the steady-state behavior of multi-bus power systems under varying operating conditions. It ensures voltage levels remain within safe limits, reduces transmission losses, and aids in optimal system planning. The results highlight that effective voltage regulation, generation scheduling, and network configuration are crucial for reliable and efficient operation. With increasing renewable integration and growing demand, load flow studies are indispensable for maintaining stability and guiding infrastructure improvements.